TILING ALMOST-SQUARES WITH SMALLER DISTINCT ALMOST-SQUARES

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Abstract. An almost-square is a $k\times(k+1)$ rectangle. We show that the $n\times(n+1)$ almost-square can be cut into almost-squares of distinct smaller sizes if and only if $n \in \{4, 10, 12, 14, 15, 18\}$ or $n \geq 20$, settling a conjecture of Erich Friedman. The construction starts from a perfect squared square, scales it, and moves each cut line by a small integer so that every square becomes an almost-square. Two classical squared squares, of sides 112 and 110, give every even and every odd n beyond an explicit bound; explicit tilings and an exhaustive search settle the rest. One counting identity proves the construction correct and also yields a parity lemma, which explains why one squared square can reach only one parity of n and another cannot be used at all.

Status of this preprint. Every finite check behind the theorem is rerun by a checker that shares no code with the programs that found the tilings (Section 6). This version has not yet been independently reviewed.

1. The question

An *almost-square* is a $k \times (k + 1)$ rectangle; call k its *size*. For which n can the $n \times (n + 1)$ almost-square be cut into almost-squares of distinct sizes, all smaller than n? Each piece may lie either way round.

The first test is area. If the sizes are $k_1, \ldots, k_m$, then

$$k_1(k_1 + 1) + \cdots + k_m(k_m + 1) = n(n + 1). \tag{1}$$

For $n = 4$ the sizes $1, 2, 3$ pass, since $2 + 6 + 12 = 20$. Before reading on, try to fit a 1×2, a 2×3 and a 3×4 into a 4×5 rectangle; one answer is on the left of Figure 2. For $n = 5$ no set of distinct sizes below 5 has total area 30, so there is nothing to try. For $n = 6$ exactly one set passes, $\{3, 5\}$, since $12 + 30 = 42$, and yet it fails. The 5×6 piece either spans the full height of the 7×6 rectangle, leaving room only two units wide, or lies flat, leaving a strip one unit thick along two sides. The 3×4 piece fits in neither. Area is necessary and far from sufficient, and Section 5 shows that the search for small n cannot be avoided.

For large n there is a better idea, and one example shows all of it. In 1925 Zbigniew Moroń published a dissection of a 33×32 rectangle into nine squares of distinct sizes [10], a *perfect squared rectangle*. That rectangle is itself an almost-square of size 32. Now move four of its vertical cut lines one unit to the left and one horizontal cut line one unit up, as in Figure 1. Each square gains or loses one unit in exactly one direction and becomes an almost-square. The new sizes $17, 15, 13, 10, 9, 8, 6, 3, 1$ are distinct, so the 33×32 almost-square has been cut into nine distinct smaller almost-squares. As a check on (1), $306+240+182+110+90+72+42+12+2 = 1056 = 32 \cdot 33$.

One tiling settles one n. To settle infinitely many we want to scale the picture, and here the rectangle lets us down: scaled by t, Moroń's rectangle becomes $33t \times 32t$, which is no longer an almost-square. A square has no such problem. Scale a squared square by t and move its cut lines so that every piece, the outer square included, gains one unit more in one direction than in the other. This works at every scale at once, and Sections 2 and 3 turn it into a proof.

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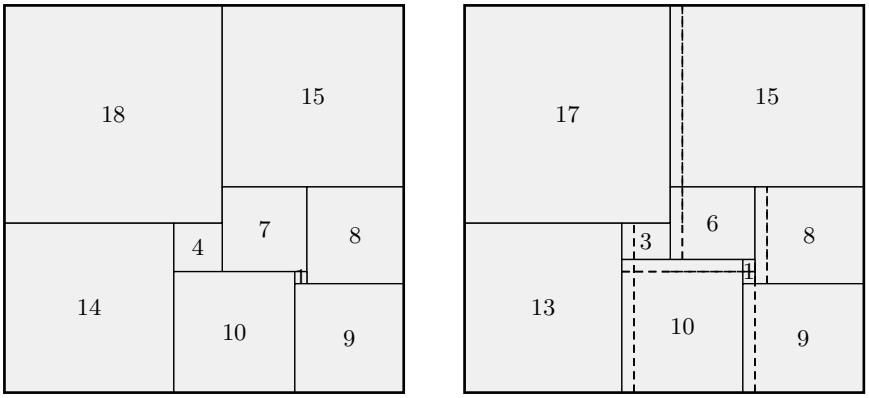


FIGURE 1. Left: Moroń's 33×32 rectangle cut into nine distinct squares, labelled by side. Right: the vertical cut lines at $x = 14, 18, 24, 25$ move one unit left and the horizontal line at $y = 10$ moves one unit up (old positions dashed). Every square becomes an almost-square, labelled by size.

The question comes from Erich Friedman's Problem of the Month for May 2012 [5]. Solvers found tilings for every n from 20 to 55 and for $n = 10, 12, 14, 15, 18$; Friedman's list of unsolved problems from the column then recorded the conjecture that every $n \geq 20$ works [7].

A neighbouring problem uses every size from 1 to m at once. The almost-square packing problem asks for the rectangle of least area into which $1 \times 2, 2 \times 3, \ldots, m \times (m+1)$ fit without overlap; Simonis and O'Sullivan [11] solved it by computer up to $m = 26$, and Ellard and MacHale [4] study packings of such rectangles. When the frame must itself be an almost-square, the area condition holds only for $m = 1, 3, 8, 20, 34$, and exact fits are known for each of them; for the last, Giovanni Resta reports finding one in January 2007 [6], and van den Berg, Braam, Moes, Suilen and Bhulai found one independently [12]. For $m > 1$ these fits are tilings in our sense, of the almost-squares of sizes 4, 15, 55 and 119. Allowing any set of distinct sizes, as Friedman does, is what makes every large n reachable.

Theorem 1. *The $n \times (n+1)$ almost-square can be cut into almost-squares of distinct sizes smaller than n if and only if $n \in \{4, 10, 12, 14, 15, 18\}$ or $n \geq 20$.*

The new parts are a construction for every $n \geq 56$, where the only earlier case we know of is $n = 119$, and the impossibility for every $n \leq 19$ outside the list. The proof has three steps. Section 2 shows that moving cut lines preserves a dissection, using one counting identity, and that one good set of moves works for all large scales. Section 3 uses the same identity to explain which squared squares can be inflated and which values of n they reach. Section 4 supplies the finite data for large n, and Section 5 settles $n \leq 19$. Section 6 describes how the finite checks were verified.

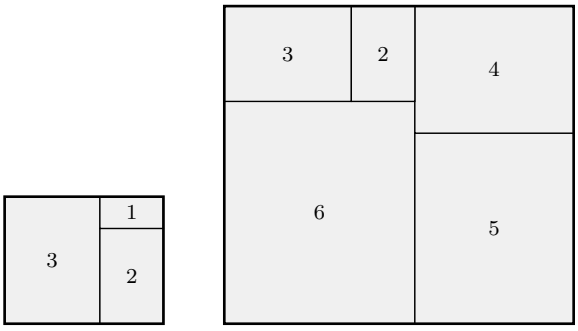


FIGURE 2. Tilings of the 5×4 almost-square by sizes $1, 2, 3$ and of the 11×10 almost-square by sizes $2, \ldots, 6$, labelled by size.

2. Moving cut lines

Throughout, coordinates, side lengths, scales and shifts are integers; Lemma 8 below shows that nothing is lost by this. Write R_n for the rectangle $[0, n+1] \times [0, n]$, the almost-square to be tiled, and call a tiling of R_n *admissible* if its tiles are almost-squares of distinct sizes, all smaller than n.

Shifts. Table 1 records Moroń's example square by square. The vertical cut lines are at $x \in \{0, 14, 18, 24, 25, 33\}$ and the horizontal ones at $y \in \{0, 9, 10, 14, 17, 32\}$. The moves are $u(14) = u(18) = u(24) = u(25) = -1$ and $v(10) = 1$, every other line staying put. The 10-square has left side at $x = 14$ and right side at $x = 24$, both moved by -1, so its width does not change; its bottom at $y = 0$ stays and its top at $y = 10$ rises by one, so it becomes 10×11.

Table 1. Moroń's squares under the moves of Figure 1. Each square is given by its side s and lower-left corner (x, y); δu and δv are the changes in its width and height. The last column is $\varepsilon = \delta u - \delta v$.

s	(x, y)	δu	δv	new shape	size	ε
18	$(0, 14)$	-1	0	17×18	17	-1
15	$(18, 17)$	1	0	16×15	15	1
14	$(0, 0)$	-1	0	13×14	13	-1
10	$(14, 0)$	0	1	10×11	10	-1
9	$(24, 0)$	1	0	10×9	9	1
8	$(25, 9)$	1	0	9×8	8	1
7	$(18, 10)$	0	-1	7×6	6	1
4	$(14, 10)$	0	-1	4×3	3	1
1	$(24, 9)$	0	1	1×2	1	-1

In general, let $Q = [0, W] \times [0, H]$ be dissected into squares $Q_i = [x_i, x_i + s_i] \times [y_i, y_i + s_i]$, $1 \le i \le m$. Let $\mathcal{X}$ be the set of x-coordinates of vertical sides of the squares and $\mathcal{Y}$ the set of y-coordinates of horizontal sides. A *shift* is a pair of integer functions u on $\mathcal{X}$ and v on $\mathcal{Y}$ with $u(0) = v(0) = 0$. At scale $t \ge 1$, put

$$X_t(a) = ta + u(a), \qquad Y_t(b) = tb + v(b),$$

so that X_t places each vertical cut line at t times its old position plus its shift, and Y_t does the same for horizontal lines. Square i becomes the rectangle

$$Q_i^t = [X_t(x_i), X_t(x_i + s_i)] \times [Y_t(y_i), Y_t(y_i + s_i)],$$

of width $ts_i + \delta u_i$ and height $ts_i + \delta v_i$, where $\delta u_i = u(x_i + s_i) - u(x_i)$ and $\delta v_i = v(y_i + s_i) - v(y_i)$ are the columns of Table 1. It is an almost-square exactly when $\delta u_i - \delta v_i = \pm 1$ (Figure 3).

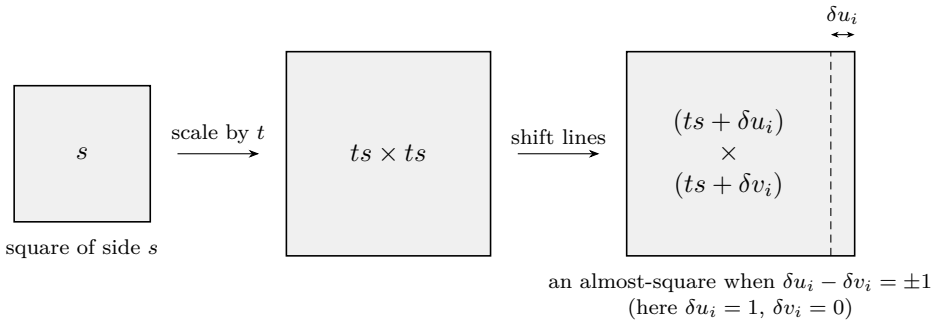


Figure 3. What a shift does to one square. After scaling, the cut lines around it move by small integers, changing its width by δu_i and its height by δv_i.

Why the pieces still fit. Moving lines could, in principle, make pieces overlap or leave gaps. Overlaps are ruled out by the order of the lines: two squares of the dissection lie on opposite sides of some cut line, and if the lines keep their order they stay on opposite sides. Gaps need a count, and the count we need is worth stating on its own, because Section 3 uses it twice more.

For Moroń's rectangle, take any function f on the vertical cut lines and any g on the horizontal ones. Each square contributes the product of how much f changes across it and how much g changes across it. With $f(x) = x$ and $g(y) = y$ the contributions are the areas s_i^2, and their sum is the area $33 \cdot 32$ of the rectangle. More surprisingly, take $f = u$ and $g(y) = y$. The contributions are $\delta u_i \, s_i$, which from Table 1 are $-18, 15, -14, 0, 9, 8, 0, 0, 0$. They add up to 0, and so does the same quantity for the whole rectangle, $u(33) \cdot 32$. The next lemma says this always happens, for every f and g.

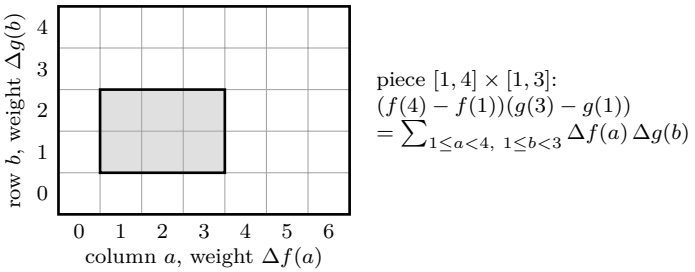


FIGURE 4. The proof of Lemma 2. Give the unit cell in column a and row b the weight $\Delta f(a)\, \Delta g(b)$. A piece's term is the total weight of its cells, and every cell lies in exactly one piece.

Lemma 2 (Balance)**.** *Let $[0, W] \times [0, H]$ be dissected into rectangles $[l_i, r_i] \times [b_i, t_i]$ with integer corners, and let f, g be integer functions on the integers. (A function given only on the cut lines can be extended to all integers in any way; only its values on the cut lines matter.) Then*

$$\sum_i \big(f(r_i) - f(l_i)\big)\big(g(t_i) - g(b_i)\big) = \big(f(W) - f(0)\big)\big(g(H) - g(0)\big).$$

The left side adds, over the pieces, the change of f across the piece times the change of g up it; the right side is the same quantity for the whole rectangle.

Proof. Write $\Delta f(a) = f(a+1) - f(a)$. The change of f across $[l, r]$ is $\sum_{l \le a < r} \Delta f(a)$, so the term for piece i is the sum of $\Delta f(a)\, \Delta g(b)$ over the unit cells $[a, a+1] \times [b, b+1]$ of that piece. Every unit cell of $[0, W] \times [0, H]$ lies in exactly one piece, so the left side is the sum of $\Delta f(a)\, \Delta g(b)$ over all unit cells, which factorises as the right side (Figure 4). □

Lemma 3 (Inflation)**.** *Suppose X_t is strictly increasing on $\mathcal{X}$ and Y_t is strictly increasing on $\mathcal{Y}$. Then the rectangles Q_i^t tile $[0, X_t(W)] \times [0, Y_t(H)]$. If moreover $\delta u_i - \delta v_i = \pm 1$ for every i, then Q_i^t is an almost-square of size $ts_i + \min(\delta u_i, \delta v_i)$.*

Proof. Two distinct squares of the dissection have disjoint interiors, so one lies entirely to the left of, or entirely below, the other; say $x_i + s_i \le x_j$. Since X_t is increasing, $X_t(x_i + s_i) \le X_t(x_j)$, and Q_i^t and Q_j^t do not overlap. Each Q_i^t lies in the big rectangle because $X_t(0) = 0$ and X_t is increasing, and likewise for Y_t. Lemma 2 with $f = X_t$ and $g = Y_t$ says that the areas of the Q_i^t add up to $X_t(W)\, Y_t(H)$, the area of the big rectangle. Non-overlapping rectangles inside it with the full total area cover every unit cell exactly once, so they tile it. The last sentence is the shape computation above. □

Squares, and every scale at once. When $W = H = S$, the big rectangle is $(tS + u(S)) \times (tS + v(S))$. It is an almost-square at every scale exactly when $u(S) - v(S) = \pm 1$; we take $u(S) - v(S) = 1$, which a reflection in the diagonal arranges. Call a shift with $\delta u_i - \delta v_i = \pm 1$ for every square and $u(S) - v(S) = 1$ an *inflation*. It turns the squared square into R_n with $n = tS + v(S)$, whose pieces have sizes $ts_i + c_i$ with $c_i = \min(\delta u_i, \delta v_i)$. Moroń's rectangle needed

$W - H = 1$ to begin with, and could not be scaled; a squared square gets its extra unit from the shift.

Our main example is the inflation of Duijvestijn's squared square of side 112 that is zero except for the values

$$u: \ 29 \mapsto -1, 65 \mapsto 1, 112 \mapsto 1$$

and

$$v: \ 33 \mapsto -1, 37 \mapsto -1, 53 \mapsto -2, 60 \mapsto -1, 62 \mapsto -1, 77 \mapsto -1.$$

Here $v(112) = 0$, so this inflation gives $n = 112t$. The 50-square in the upper-left corner has $\delta u = 0$ and $\delta v = v(112) - v(62) = 1$; at $t = 2$ it becomes 100×101, the almost-square of size 100. Every c_i is 0 or -1. At $t = 1$ two sizes clash: the 16-square has $c = -1$ and the 15-square has $c = 0$, giving sizes $16t - 1$ and $15t$, which agree only at $t = 1$. From $t = 2$ on the 21 sizes are distinct, and Figure 5 shows the tiling of R_{224}.

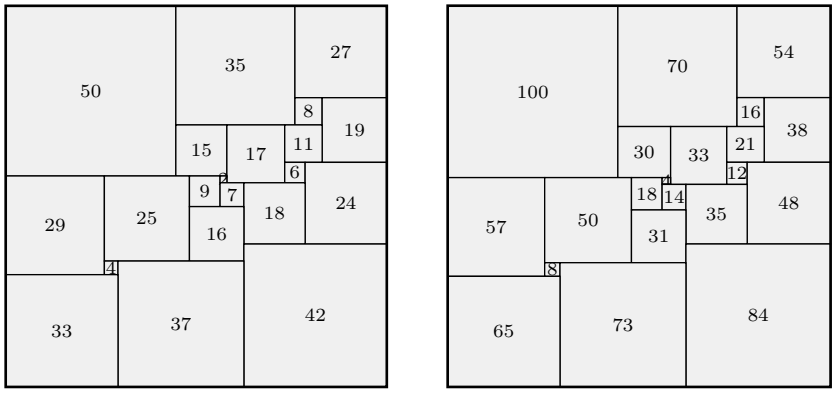


FIGURE 5. Duijvestijn's squared square of side 112 and its inflation at $t = 2$, an admissible tiling of R_{224} by 21 almost-squares. Three vertical and six horizontal cut lines have moved, by at most two units.

The clash is the only thing that can go wrong for large t, and it can go wrong at most once for each pair of squares.

Lemma 4 (All large scales)**.** *Let u, v be an inflation of a squared square of side S whose squares have distinct sides, and put $n(t) = tS + v(S)$. Let t_0 be the least positive integer that exceeds every integer collision point $(c_j - c_i)/(s_i - s_j)$ and at which every consecutive gap $t(a' - a) + u(a') - u(a)$ and $t(b' - b) + v(b') - v(b)$, every size $ts_i + c_i$ and every difference $n(t) - (ts_i + c_i)$ is positive. Then for every $t \geq t_0$, Lemma 3 gives an admissible tiling of $R_{n(t)}$.*

Proof. Each quantity listed is an affine function of t with positive slope, so once positive it stays positive. Two sizes $ts_i + c_i$ and $ts_j + c_j$ with $s_i \neq s_j$ agree for at most one t, the collision point. For $t \geq t_0$ the cut lines keep their order, every tile is an almost-square of positive size less than $n(t)$, and the sizes are distinct. □

3. Which squared squares inflate

The example above reaches $n = 112t$. Can some other inflation of the same square reach an odd n? That needs $v(112)$ odd, and the answer is no, for a reason that can be seen in six squares.

Figure 6 shades a stack of six squares of the 112-square, of sides $42, 18, 6, 11, 8, 27$ from the bottom edge to the top edge. Each square's top is the next square's bottom, so the height changes along the stack telescope: $\sum \delta v_i = v(112) - v(0) = v(112)$. The widths behave differently. The squares run between $x = 70$ and 112, 70 and 88, 82 and 88, 82 and 93, 85 and 93, and 85 and 112, so

$$\sum \delta u_i = 2\big(u(112) + u(88) + u(93) - u(70) - u(82) - u(85)\big),$$

an even number. Now each $\varepsilon_i = \delta u_i - \delta v_i$ is ± 1, and a sum of six odd numbers is even. Hence $v(112) = \sum \delta u_i - \sum \varepsilon_i$ is even, for every inflation. In the example the ε_i are $1, 1, -1, 1, -1, 1$, $\sum \delta u_i = 2$ and $v(112) = 0$.

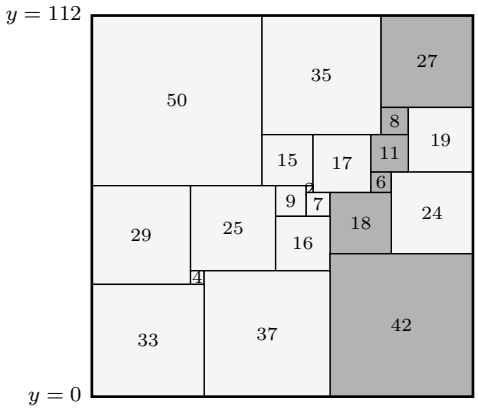


FIGURE 6. Six squares of Duijvestijn's square stacked from bottom to top. Each interior horizontal line between them is the top of one and the bottom of the next, and each of $x = 70, 82, 85, 88, 93, 112$ is a vertical side exactly twice. Only $y = 0$ and $y = 112$ occur once.

Nothing in this argument used the stack shape. It used only that each x-coordinate occurs an even number of times among the vertical sides of the chosen squares, and each y-coordinate among their horizontal sides, apart from the two we wanted to learn about.

Lemma 5 (Parity)**.** *Let u, v be a shift with $\delta u_i - \delta v_i = \pm 1$ for all i, and let T be a set of squares. Then*

$$\sum_{i \in T} (u(x_i) + u(x_i + s_i) + v(y_i) + v(y_i + s_i)) \equiv |T| \pmod 2.$$

In particular, if every x-coordinate occurs an even number of times among the vertical sides of the squares in T, and every y-coordinate among their horizontal sides, then $|T|$ is even.

Proof. For each i, $\delta u_i - \delta v_i$ is odd and is congruent to $u(x_i) + u(x_i + s_i) + v(y_i) + v(y_i + s_i)$ modulo 2, because $-1 \equiv 1$. Sum over T. If every coordinate occurs an even number of times, every shift value appears an even number of times on the left, so the left side is even. □

Squared squares are named below by their Bouwkamp codes, as listed by Moews [9]; each square is written $(s; x, y)$ with lower-left corner (x, y). The square of side 112 is Duijvestijn's, of order 21, the lowest possible [2]. Squares of side 110 were found by Duijvestijn [3], and Gambini showed that 110 is the smallest side of any perfect squared square [8].

Proposition 6.

1. *Every inflation of Duijvestijn's square of side* 112*, code* $(50, 35, 27)(8, 19)\ldots$*, has* $v(112)$ *even.*
2. *Every inflation of the order-*22 *square of side* 110 *with code* $(60, 50)(27, 23)\ldots$ *has* $v(110)$ *odd.*
3. *The order-*22 *square of side* 139 *with code* $(80, 59)(21, 38)\ldots$ *has no inflation.*

Proof. Apply Lemma 5 with the following sets.

(1) Take the stack of Figure 6,

$$T = \{(42; 70, 0), (18; 70, 42), (6; 82, 60), (11; 82, 66), (8; 85, 77), (27; 85, 85)\}.$$

Every coordinate occurs an even number of times except $y = 0$ and $y = 112$, so $v(0) + v(112) \equiv 6 \equiv 0$.

(2) Take

$$T = \{(26;0,0),(22;24,28),(18;92,0),(16;75,17),(14;46,36),$$
$$(12;63,21),(9;54,21),(6;54,30),(2;24,26),(1;91,17)\}.$$

Every coordinate occurs an even number of times except $x = 0$ and $x = 110$, so $u(110) \equiv 10 \equiv 0$, and $v(110) = u(110) - 1$ is odd.

(3) Take

$$T = \{(80;0,59),(59;80,80),(38;101,42),(32;61,0),(30;0,0),(28;29,31),(24;93,0),$$
$$(22;117,0),(20;119,22),(18;101,24),(7;57,35),(3;61,32),(1;29,30)\}.$$

It has thirteen squares and every coordinate occurs an even number of times, so 13 would be even. □

The signed identity. Lemma 2 says more. In Moroń's example the ε column of Table 1 splits the sides into $18 + 14 + 10 + 1 = 43$ and $15 + 9 + 8 + 7 + 4 = 43$, two equal halves. For the side-112 inflation the split is 273 against 161, which differ by 112.

Corollary 7. *For a shift of a dissection of $[0, W] \times [0, H]$ into squares, $\sum_i \delta u_i\, s_i = H\, u(W)$ and $\sum_i \delta v_i\, s_i = W\, v(H)$. For an inflation of a squared square of side S, with $\varepsilon_i = \delta u_i - \delta v_i$,*

$$\sum_i \varepsilon_i\, s_i = S.$$

Proof. Lemma 2 with $f = u$, $g(y) = y$ gives the first identity, and with $f(x) = x$, $g = v$ the second. Subtract, with $W = H = S$ and $u(S) - v(S) = 1$. □

So an inflation chooses, for each square, which way round its almost-square lies, in such a way that the signed sides add up to S. In Moroń's example $W = 33$, $H = 32$ and $u(33) = v(32) = 0$, and both identities give 0. This is the electrical balance of Brooks, Smith, Stone and Tutte [1] in another guise: along any cut line, the squares on either side have the same total length.

4. Large n

Let Σ_{112} and Σ_{110} be the squares of Proposition 6(1) and (2). For each even residue r modulo 112 there is an inflation of Σ_{112} with $v(112) \equiv r$, and for each odd residue r modulo 110 an inflation of Σ_{110} with $v(110) \equiv r$; the example of Section 2 is the class $r = 0$. That makes $56 + 55 = 111$ inflations, one per class, found by a constraint solver and listed in the ancillary data with their thresholds t_0 from Lemma 4. Each of these 111 has threshold at most 3 and reaches its first n by 360; two of the spares have $t_0 = 4$. By Lemma 4, every $n = tS + v(S)$ with $t \geq t_0$ has an admissible tiling. Proposition 6 shows that the parities cannot be swapped between the two squares.

These classes leave exactly 151 values of n with $20 \leq n \leq 248$ uncovered, and every $n \geq 249$ is covered. For each uncovered n the data contain an explicit admissible tiling. Those with $n \leq 45$ came from a direct search. The rest came from Lemma 3 at a fixed scale $t \leq 3$, applied to perfect squared rectangles from Moews's catalogue of orders up to 16 [9] whose sides are nearly equal: a scaled rectangle $tW \times tH$ with $tW - tH$ small can still be shifted to an almost-square at that one scale. Moroń's rectangle gives $n = 60$ and $n = 98$ at $t = 2$ and $t = 3$ this way, and a single 57×55 rectangle of order 10 gives 67 of the 151 cases at $t = 1$. At one fixed scale the shifts need not be small, only order-preserving: for $n = 99$ the outer sides of the rectangle move by more than 40 units, and different shifts of the same rectangle reach different n. Figure 7 shows how the range $20 \leq n < 260$ is shared out.

5. Small n

For $n \leq 19$ there is no construction to find, only a search to run. The area test alone does little, as Table 2 shows: from $n = 6$ on, every n has a set of sizes with the right area, and the number of such sets grows quickly.

Before searching we need one fact. A tiling might, in principle, place tiles at fractional positions, and a search over integer positions would then miss it. It cannot happen.

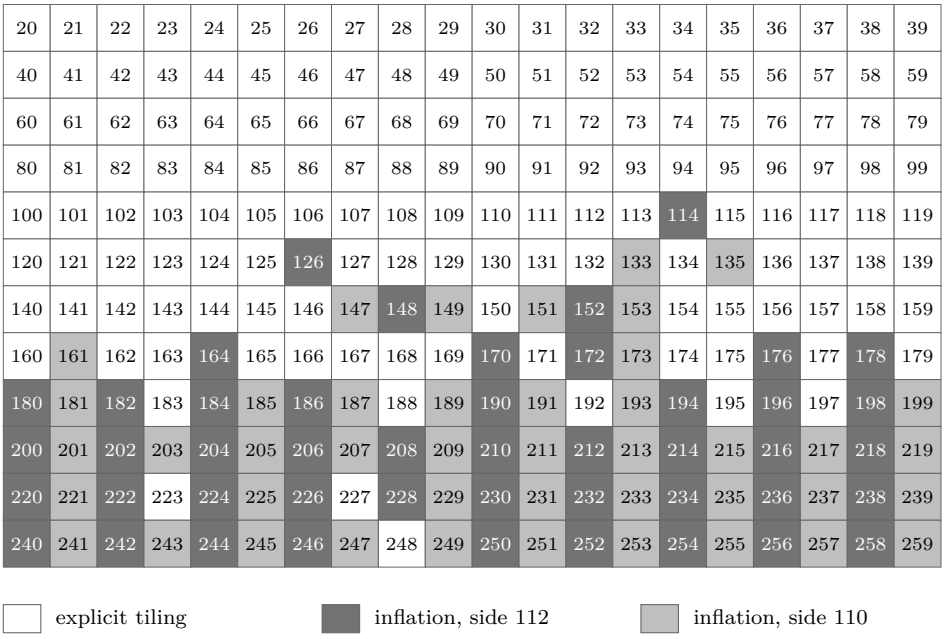


FIGURE 7. How each n with $20 \le n < 260$ is tiled: by an inflation of the square of side 112 (even n, dark) or 110 (odd n, light), or by one of the 151 explicit tilings (white). Every $n \ge 249$ is covered by an inflation.

TABLE 2. For each $n \le 19$, the number of sets of distinct sizes below n satisfying the area equation (1), and whether an admissible tiling exists.

n	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
area sets	0	0	1	0	1	2	1	3	6	5	3	13	14	18	30	27	42	56
tiling			yes						yes		yes		yes	yes			yes	

Lemma 8 (Integer corners)**.** *Let $[0, W] \times [0, H]$, with W and H integers, be the union of finitely many closed rectangles with integer side lengths and pairwise disjoint interiors. Then every corner of every rectangle has integer coordinates.*

Proof. Let A be the finite set of x-coordinates of the vertical sides of the rectangles, and suppose some element of A is not an integer. Let a be the least such element. It is the left side of some rectangle P: if it were a right side, the left side $a - w$ of the same rectangle would be a smaller non-integer in A. Also $a > 0$, since 0 is an integer.

Let c be the largest element of $A \cup \{0\}$ below a; it is an integer. Pick a height y_0 strictly inside the vertical extent of P that is not the y-coordinate of any horizontal side. The point $((c + a)/2, y_0)$ lies in the big rectangle, so some rectangle $P' = [l, r] \times [\,\cdot\,]$ contains it. Then $l < a$, so $l \le c$, and $r > c$, so $r \ge a$, since no element of A lies strictly between c and a. If $r > a$, then points just to the right of (a, y_0) lie in the interiors of both P and P' (here y_0 is not on any horizontal side), which is impossible. So $r = a$, and $a = l + (r - l)$ is the sum of an integer, being an element of A below a, and an integer width. This contradicts the choice of a (Figure 8). The same argument applies to y-coordinates. □

So it suffices to search over integer placements. The search fills R_n from the bottom up. In any partial tiling, the filled region is bounded above by a skyline. Take a segment of the skyline that is a local minimum, with a higher wall (or the boundary of R_n) on each side; the program takes the narrowest such segment. The cell at its left end has filled cells below it and to its left, so some tile has its lower-left corner exactly there, and the wall on the right bounds that tile's width. It suffices to try each unused size in each orientation at that point. (The lowest, leftmost unfilled cell is one such point; the Lean search uses that one.) The search finds tilings exactly

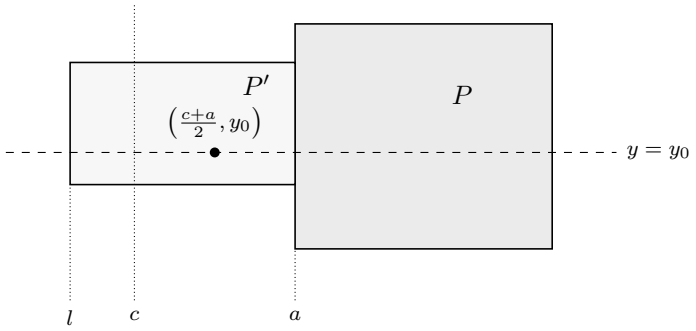


FIGURE 8. Lemma 8. The rectangle P' covering the point between c and a must end exactly at a; otherwise it would overlap P along the line $y = y_0$. Its left side l is an integer, so a is too.

for $n \in \{4, 10, 12, 14, 15, 18\}$, agreeing with Friedman's table from $n = 6$, and none for the other $n \leq 19$. The hardest case, $n = 19$, takes about 2.3×10^5 search nodes. A second program sharing no code with the first encodes the question as an exact cover (one variable per placement of each size and orientation, every cell covered once, every size used at most once, symmetries of the rectangle broken at its corners) and hands it to a satisfiability solver. It reaches the same verdict for every $n \leq 19$. Together with Section 4, this proves Theorem 1. □

6. Verification

Every claim in Sections 4 and 5 is a finite check, and the data and programs are at `https://github.com/jvvk/mathematics/tree/main/almost-square-tilings`: the layouts of the squared squares, the 128 inflations with their thresholds (the 111 used and some spares), the 151 explicit tilings and the six small ones, both searches for $n \leq 19$, checksums, and a script that reruns the checker, the search in C for $n \leq 19$, the second search for $n \leq 18$ ($n = 19$ on request, a few minutes) and the checksums. A checker sharing no code with the programs that found the tilings rebuilds each tiling cell by cell; it checks every explicit tiling, every inflation at its first two scales and the conditions of Lemma 4 for 300 further scales, and confirms that every n with $20 \leq n \leq 20000$ is covered. To make sure the checker is not vacuous, the certificate was corrupted in four ways (moving one tile, deleting a tiling, changing one shift value, removing a residue class), and each corruption was reported. The checker also caught a real error in an early certificate, a threshold computed without the collision points of Lemma 4.

7. Further questions

Question 9. For which perfect squared squares is the parity obstruction of Lemma 5 the only obstruction to an inflation? For the three squares of Proposition 6, reduction modulo 2 already decides every question asked.

Question 10. The same construction applies to $k \times (k + c)$ rectangles for fixed $c \geq 1$, the condition becoming $\delta u_i - \delta v_i = \pm c$. For which n can the $n \times (n + c)$ rectangle be cut into distinct smaller $k \times (k + c)$ rectangles?

Question 11. What is the least number of pieces in an admissible tiling of R_n? The inflations use 21 or 22 pieces for every large n, and Moroń's example uses nine for $n = 32$.

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